

HMMT 2014
Saturday 22 February 2014

HMIC

1. [20] Consider a regular n -gon with $n > 3$, and call a line *acceptable* if it passes through the interior of this n -gon. Draw m different acceptable lines, so that the n -gon is divided into several smaller polygons.
 - (a) Prove that there exists an m , depending only on n , such that any collection of m acceptable lines results in one of the smaller polygons having 3 or 4 sides.
 - (b) Find the smallest possible m which guarantees that at least one of the smaller polygons will have 3 or 4 sides.
2. [25] 2014 triangles have non-overlapping interiors contained in a circle of radius 1. What is the largest possible value of the sum of their areas?
3. [30] Fix positive integers m and n . Suppose that $a_1, a_2, \dots, a_m$ are reals, and that pairwise distinct vectors $v_1, \dots, v_m \in \mathbb{R}^n$ satisfy

$$\sum_{j \neq i} a_j \frac{v_j - v_i}{\|v_j - v_i\|^3} = 0$$

for $i = 1, 2, \dots, m$.

Prove that

$$\sum_{1 \leq i < j \leq m} \frac{a_i a_j}{\|v_j - v_i\|} = 0.$$

4. [35] Let ω be a root of unity and f be a polynomial with integer coefficients. Show that if $|f(\omega)| = 1$, then $f(\omega)$ is also a root of unity.
5. [40] Let n be a positive integer, and let A and B be $n \times n$ matrices with complex entries such that $A^2 = B^2$. Show that there exists an $n \times n$ invertible matrix S with complex entries that satisfies $S(AB - BA) = (BA - AB)S$.